

C++

PENG

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1 Introduction

c++ is a high-level, statically typed programming language [1].

1.1 IO

Definition 1.1 (IO objects).

- `iostream` : *std lib*
- `std::cin` : *istream type*
- `std::cout` : *ostream type*

Definition 1.2 (namespace).

- `std` : *namespace*

- `::` : *scope operator*

```
std::cout << "str";           // return object: std::cout
std::cout << "str" << "\n"; // (std::cout << "str") << "\n";

int v1 = 0, v2 = 0;
std::cin >> v1;               // return object: std::cin
std::cin >> v1 >> v2;         // (std::cin >> v1) >> v2;
```

1.2 Comments

Definition 1.3 (comment).

```
// single line, best way
/* paired, can not nest */

std::cout << /* "/" /* "/" */;
// remove everything between */ and closest /* before
// std::cout << " /* ";
```

1.3 Flow of Control

Definition 1.4 (while).

```
while(cond)
{
    statements;
}

while(std::cin >> val)
// std::cin >> val return left operand: std::cin
// istream become invalid:
// 1. reading a non-integer val: 1 2 3.1
// 2. end-of-file: Ctrl-c
```

Definition 1.5 (for).

```
for (init-statement; cond; expr)
{
    statements;
}

// init-statement:
// local var used only inside for, executed only once
// expr: executed after for body
```

Definition 1.6 (if).

```
if (cond)
{
    statements;
}
else if (cond)
{
    statements;
}
else
{
    statements;
}
```

```
if (std::cin >> val)
    std::cout << val << "\n";
// 1 2 3 RET => 1
```

1.4 Class

Definition 1.7 (class). *Define data structures:*

- *name*
- *defined*
- *operation*

```
// #include <std lib>
// #include "header.h"
```

Definition 1.8 (member function).

```
data.func()
```

2 Variables and Basic Types

2.1 Primitive Built-in Types

2.1.1 Arithmetic Types

Definition 2.1 (Primitive builtin types).

- *arithmetic types:*
 - *integral types*
 - * *characters:* `char`, `std::string`
 - * *integers:* `int`: signed or unsigned (≥ 0)
 - * *boolean values:* `bool`
 - *floating-point types:* `double`
- *void*

Remark 2.2. Use `double` for floating-point computations.

2.1.2 Type Conversions

Example 2.3 (Type conversions).

```
bool b = 20; // 0 for false and nonzero values for true
int i = b;   // true for 1 and false for 0
```

Example 2.4 (Signed and unsigned).

```
unsigned u = -1; // if 32-bit ints, 4294967295
int a = -1;
unsigned b = 1;
a * b // int value is converted to unsigned before addition
```

2.1.3 Literals

Definition 2.5 (Literals).

Integer literals:

- *decimal*
- *octal*: 0-
- *hexadecimal*: 0x-
- *suffix*:
 - *u*: *unsigned*
 - *L*: *long*

Floating-point literals (double by default):

- *decimal point*
- *E*: 1E-2
- *suffix*: *f*/*F*: *float*

character and string literals: "Hello" = "Hello\0"

Definition 2.6 (Escape sequences).

- *newline*: \n
- *horizontal tab*: \t
- *null*: \0

2.2 Variables

2.2.1 Variable Definitions

Remark 2.7.

- *Variables defined outside any function body are initialized to 0.*
- *Initialize every object of built-in type.*
- `std::cin >> int i;` *error, should be* `int i; std::cin >> i;`.

2.2.2 Variable Declarations

- File A define the variable `int ix;`. Other files: `extern int ix;`.
- All variables must be declared before being used.

2.2.3 Identifiers

Naming conventions:

- Variable name: lowercase
- Class name: uppercase

2.2.4 Scope of a Name

- Global scope: `::global_var`
- Block scope

Remark 2.8. *It is almost always a bad idea to define a local variable with the same name as a global variable that the function uses.*

2.3 Compound Types

2.3.1 References

Definition 2.9 (Reference). *A reference defines an alternative name for an object.*

```
int ival = 0;
int &refVal = ival;
```

- A reference is an alias, not an object.
- The type of a reference and the object must match exactly.
- A reference can only refer to the object to which it was initially bound.

2.3.2 Pointers

Definition 2.10 (Pointer). *A pointer holds the address of another object.*

```
int ival = 0;
int *pi = &ival;
*pi = 10;
int* p1, p2; // legal, but might be misleading. p2 is int.
```

- A pointer is an object. The type of the pointer and the object to which it points must match.
- Dereferencing a pointer yields the object to which the pointer points.
- States of address value stored in a pointer:
 - point to an object
 - null pointer, not bound to any object: `nullptr`

Remark 2.11 (Initialize all pointers). *Define a pointer only after the object to which it should point, or initialize the pointer to `nullptr`.*

Example 2.12 (Pointers to pointers).

```
int ival = 1;
int *pi = &ival;
int **ppi = &pi;
```

Example 2.13 (References to pointers).

```
int i = 1;
int *p;
int *&r = p;
```

2.4 Qualifier

Definition 2.14. `const`:

- *Make a variable unchangeable.*
- *Must be initialized.*

Example 2.15. *Share a `const` object among multiple files:*

```
extern const i = 10; // file A
extern const i; // other files
```

2.4.1 Reference to const

Definition 2.16 (reference to const). *Reference to const only restricts what we can do through that reference.*

Reference to const can refer to a nonconst object.

```
const int ci = 10;
const int &r1 = ci;
int i = 1;
const int &r2 = i;
const int &r3 = 2;
double d = 3.14;
const int &r4 = d;
```

2.4.2 Pointer to const

Definition 2.17 (Pointer to const: low-level). *Pointer to const only affects what we can do with the pointer.*

Pointer to const can point to a nonconst object.

```
double pi = 3.14;
const double *ptr = &pi;
```

2.4.3 Top-Level const

Definition 2.18 (const pointer: top-level). *const pointer must be initialized.*

```
int i = 10;
int *const pi = &i;
```

Remark 2.19. *Low-level const is never ignored when copying an object. Both objects must have the same low-level qualification or binding const to nonconst.*

```
const int i = 1;
const int *p1 = &i;
const int *const p2 = p1;
int *p3 = 0;
p1 = p3;
```

2.4.4 Constant Expressions

Definition 2.20 (constexpr).

```
const int sz = get_size(); // not const expression
constexpr int sz = size(); // ok only if size is a constexpr function
const int *p = 0;          // pointer to const int
constexpr int *q = 0;      // const pointer to int: top-level const
```

Remark 2.21.

- *Variables defined inside a function are not stored at a fixed address.*
- *The address of an object defined outside of any function is a constant expression.*

2.5 Dealing with Types

2.5.1 Type Aliases

Definition 2.22 (type alias). `typedef double wages;`
`typedef wages base, *p; // p for double*`
`base d = 3.14;`
`p p1 = &d, p2 = &d;`

Definition 2.23 (pointer, const, type alias).

```
typedef char *pstring; // char*
const pstring cstr = 0; // char *const
const pstring *ps;      // char **const
```

2.5.2 The Auto Type Specifier

Definition 2.24 (auto). *A variable that uses auto must have an initializer.*

The initializers for all the variables in the declaration must have consistent types.

```
auto i = 0, *p = &i;
auto sz = 0, pi = 3.14;
```

Example 2.25 (compound types, const, and auto).

```
int i = 1;
const int ci = i;
auto a = ci;      // int, top level const is dropped
const auto b = ci; // const int
auto c = &i;      // int*
auto d = &ci;     // const int* (& of a const object is low-level const)
```

3 Defining Out Own Data Structures

```
struct Node
{
    int val;
    Node *next;
    Node(int x) : val(x), next(nullptr) {}
};

int main()
{
    Node *n1 = new Node(1);
    Node *n2 = new Node(2);
    n1->next = n2;
    while (n1 != nullptr)
    {
        std::cout << n1->val << std::endl;
        n1 = n1->next;
    }
    return 0;
}
```

References

- [1] S.B. Lippman, J. Lajoie, and B.E. Moo. *C++ Primer*. Pearson Education, 2012. ISBN: 9780133053036.